

HYGIENE, DISPOSABLE INTERFACES, AND STERILITY EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device incorporates hygiene and sterility features intended to support safe repeated use, simplified cleaning procedures, and optional single-use configurations. Hygiene systems may be implemented through material selection, removable interfaces, disposable components, or protective barrier structures.

The insertable measurement section may be constructed from body-safe materials resistant to fluid absorption, microbial retention, and chemical degradation. Suitable materials may include medical-grade silicones, elastomers, thermoplastic polymers, coated surfaces, or antimicrobial-treated materials configured to withstand repeated cleaning cycles.

In certain embodiments, the device may utilize removable disposable sleeves configured to cover all or part of the insertable section during operation. Disposable sleeves may be formed from thin elastomeric or polymeric materials that maintain sensor functionality while providing hygienic separation between the device and the user.

Disposable interfaces may include replaceable distal caps, inflatable cuff covers, or modular sensing sleeves that are removed after each session. Such components may be single-use, limited-use, or sterilizable depending on implementation requirements and regulatory classification.

The device may further support sealed sensor windows and protected sensing cavities designed to prevent fluid ingress while maintaining optical, acoustic, or pressure sensing performance. Sealing structures may include bonded membranes, compliant gaskets, overmolded encapsulation layers, or integrated protective coatings.

In certain embodiments, cleaning workflows may be guided through associated software applications. Users may receive step-by-step cleaning instructions, drying recommendations, or verification prompts confirming completion of maintenance procedures prior to subsequent use.

The handle portion and non-insertable components may be constructed with smooth external surfaces, minimal crevices, and sealed interfaces to allow wipe-down sanitation using approved cleaning agents. Electrical and communication ports may be protected through sealed connectors or wireless charging architectures to reduce contamination risk.

Sterility embodiments may include pre-packaged sterile versions of the device, sterile disposable insertable modules, or sterilization-compatible constructions allowing use in clinical or research environments. Devices may be compatible with common sterilization approaches such as chemical disinfection, ultraviolet exposure, or low-temperature sterilization methods depending on material selection.

In certain implementations, hygiene monitoring may be integrated into device operation. Software systems may track usage count, cleaning intervals, or disposable component replacement schedules and may restrict operation if maintenance requirements are not met.

Accordingly, the disclosed hygiene and sterility embodiments provide layered protection supporting consumer wellness use, shared research deployment, clinical-support workflows, and future regulated applications while maintaining measurement reliability and user safety.